Day 12 Assignment

Description:

This project implements a modern Learning Management System (LMS) interface named **Sarva-Gyani**, developed using React and designed to demonstrate advanced React concepts while simulating a student learning dashboard.

The platform provides a structured environment where a student can view recent courses, track learning tasks with deadlines, and navigate through course modules. Advanced React features such as **Lazy Loading and Code Splitting** are used to dynamically load course modules only when required, improving performance and reducing initial load time.

Pure Components are used to optimize rendering and prevent unnecessary UI updates. The system also incorporates **Error Boundaries** to ensure that component failures do not crash the entire application, displaying a fallback UI instead. Additionally, **React Portals** are implemented to display notifications and modal dialogs above the main interface.

Overview:

User Story 1 – Lazy Loading & Code Splitting:
Course modules are dynamically loaded only when the user opens a course, reducing the initial bundle size and improving application performance.

User Story 2 – Pure Components:
Dashboard statistics widgets are optimized using pure components so they re-render only when their data changes.

User Story 3 – Error Boundary:
The application safely handles component errors by displaying a fallback UI instead of crashing the entire interface.

User Story 4 – React Portals:
Notifications and modal popups are rendered outside the main DOM hierarchy to ensure they appear above all interface elements.

Key Inclusions:

- Lazy Loading using **React.lazy()**
- Code Splitting with **dynamic component imports**
- **Suspense fallback loader** during component loading
- Performance optimization using **React.memo (Pure Components)**
- **Selective component re-rendering** demonstration
- **Error Boundary implementation** using getDerivedStateFromError() and componentDidCatch()
- Friendly **fallback UI for component failures**
- **React Portal implementation** using ReactDOM.createPortal()
- Modal popup and notification system
- Professional **LMS-style dashboard UI** with courses and tasks
- Modular React component architecture

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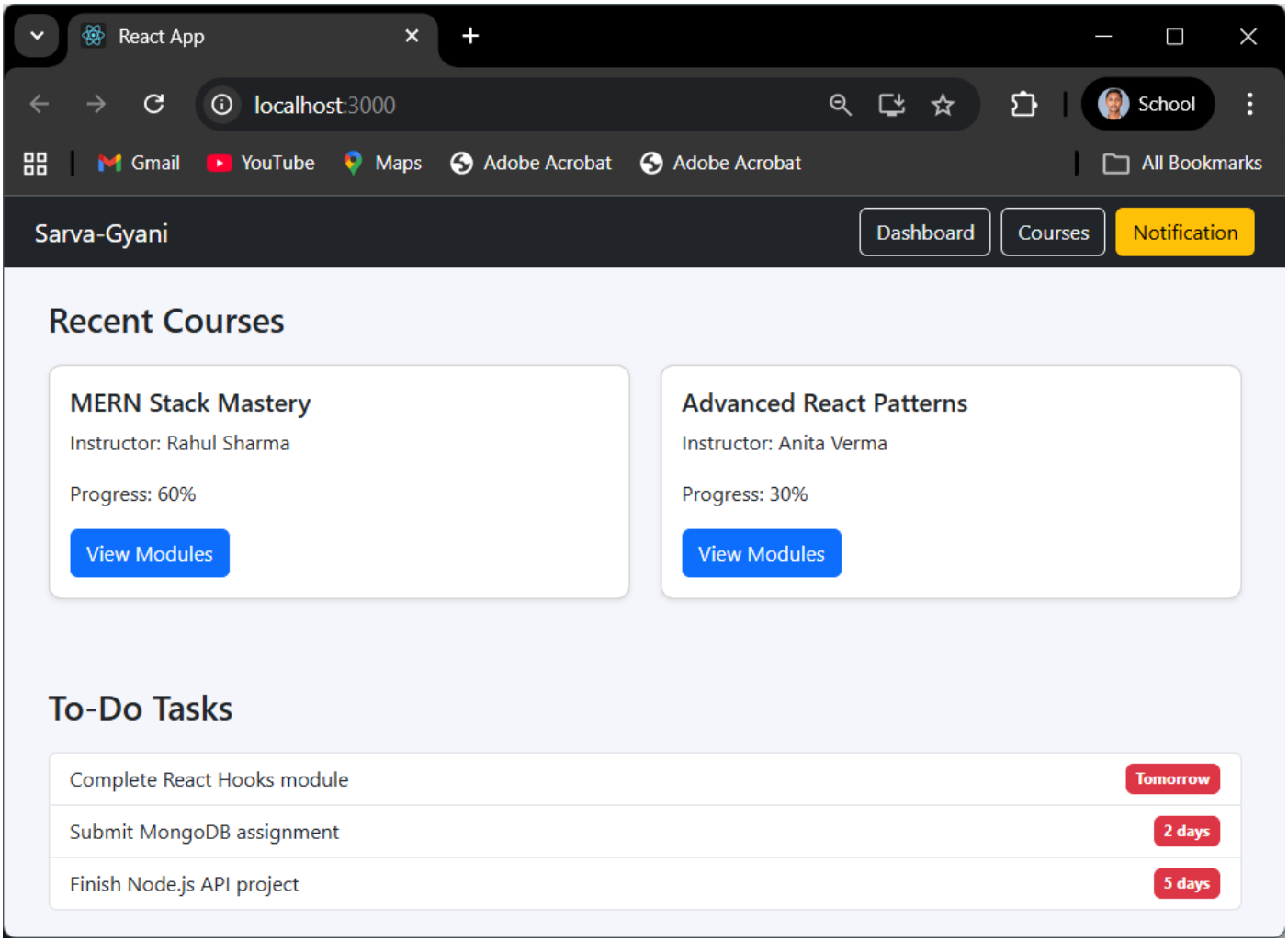
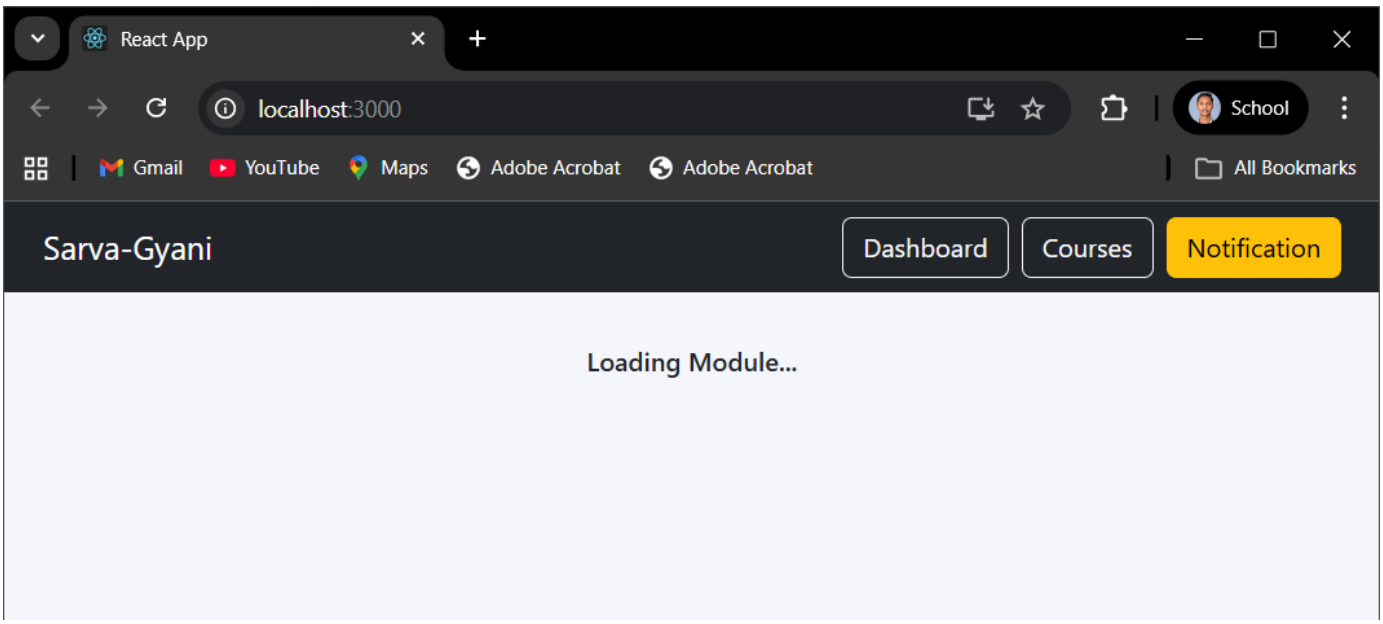
Screenshots:

Dashboard (With a Task Notifier – To Do):



Lazy Loading:



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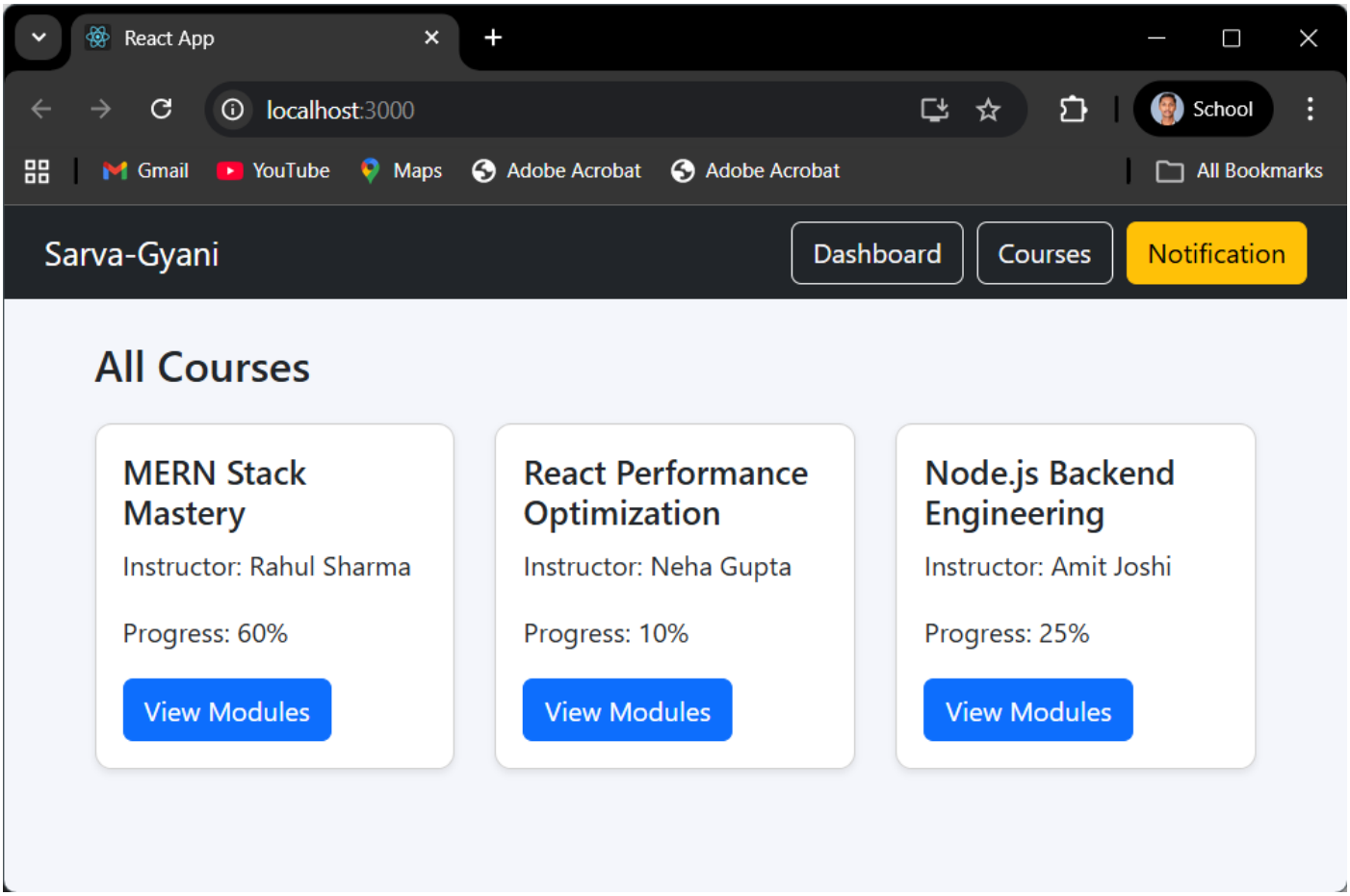
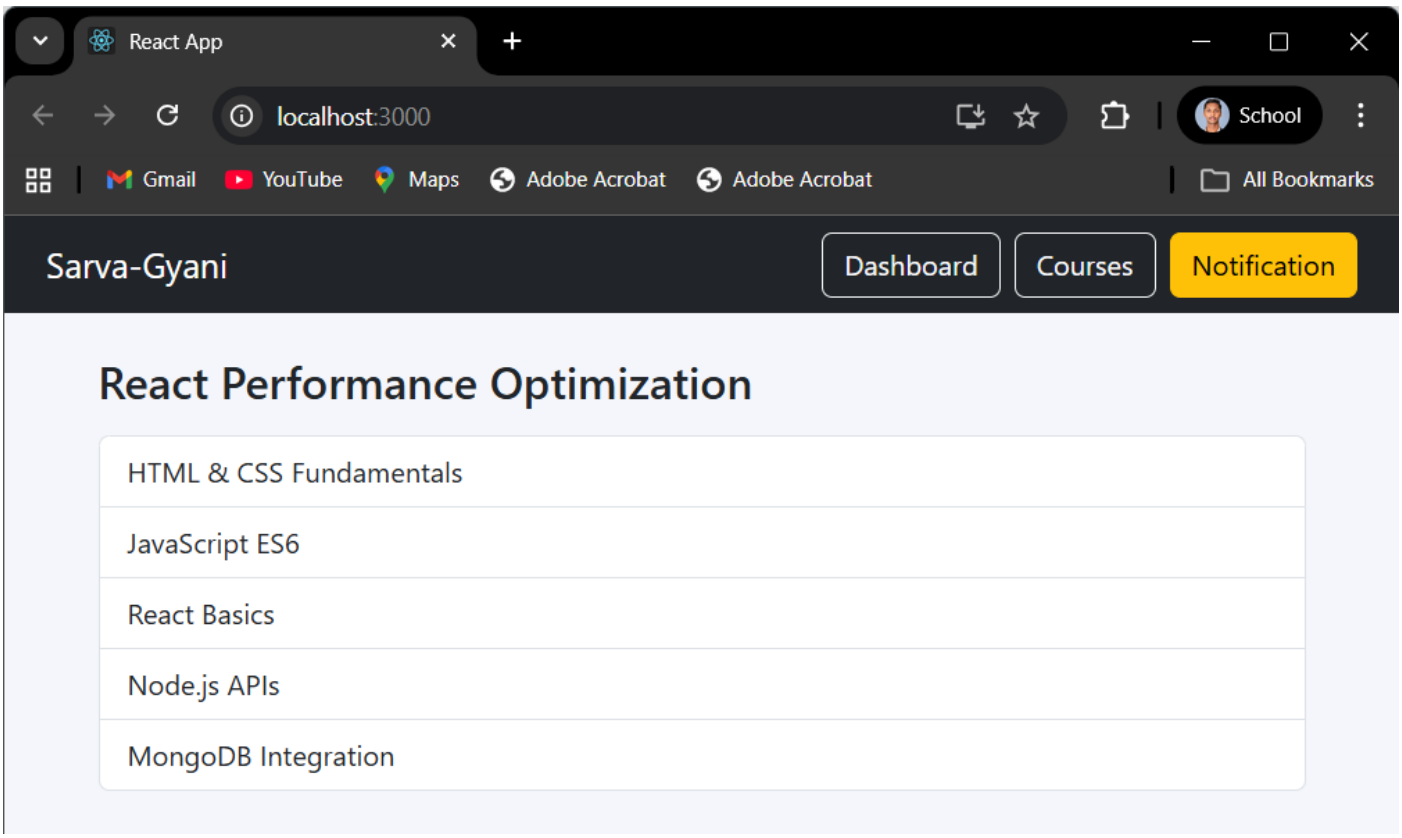
Courses:



Modules in Course:



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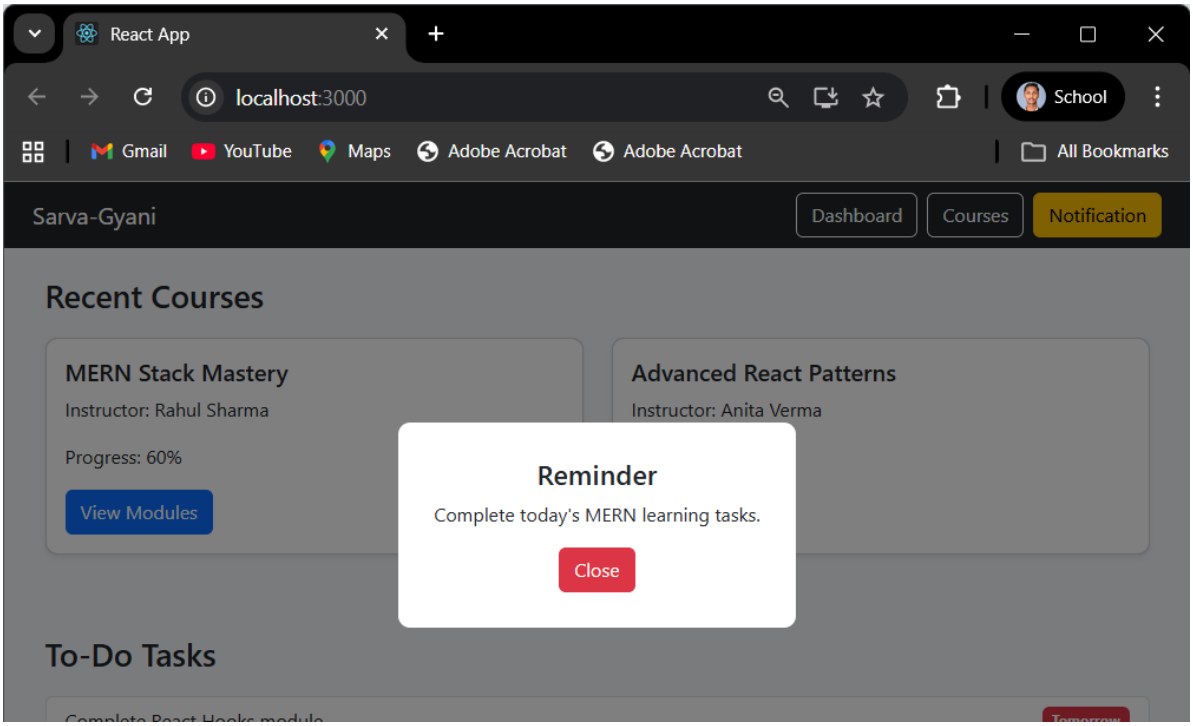
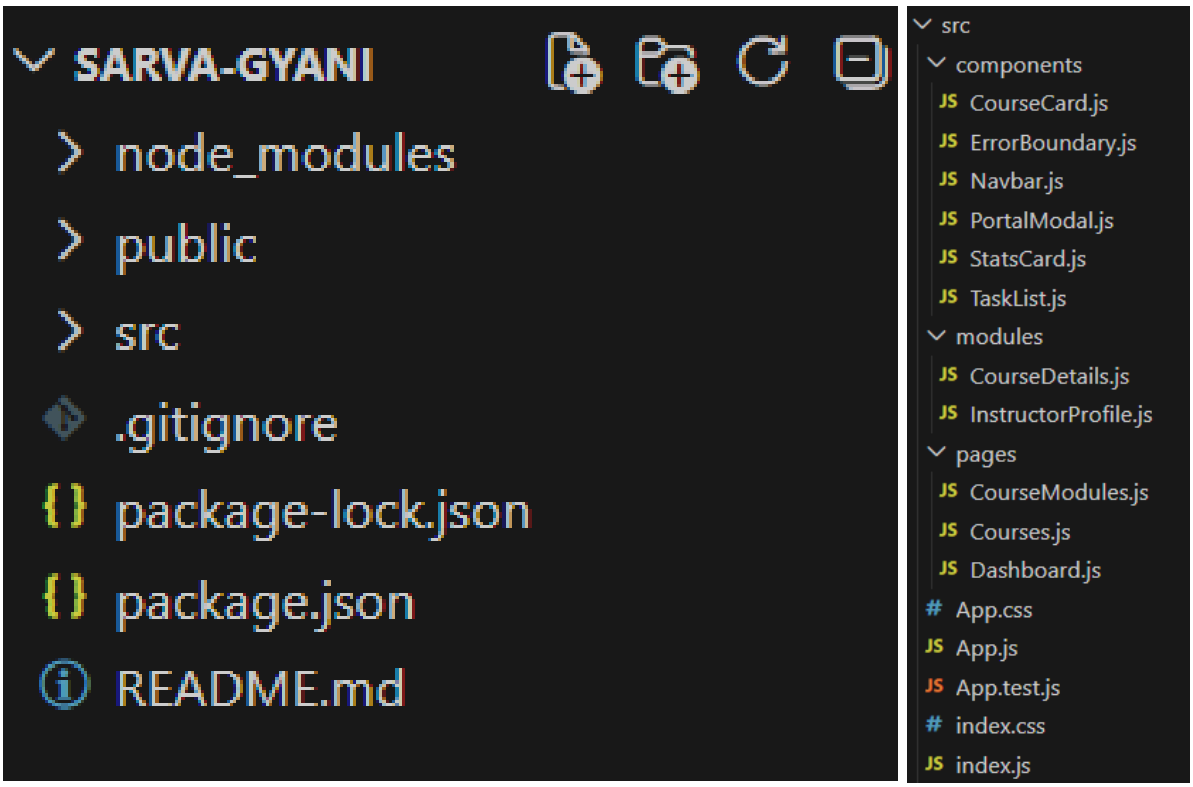
Notification Popup:



Folder Structure:



Note: The ZIP file attached in the submission does not contain the ***node_modules*** folder. Kindly create a React application using the command ***"npx create-react-app sarva-gyani"***, and then replace the **src** folders with the ones provided in the ZIP file.

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